

27 September 2026 11:22

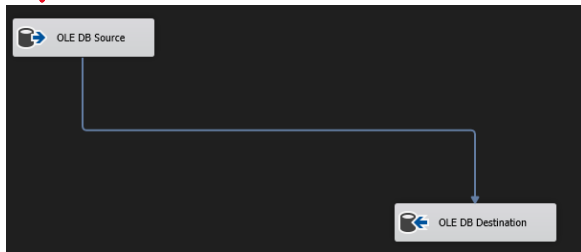
Building the ETL Pipeline in SSIS (Visual Studio)

Before I start I installed **SQL Server Integration Services Projects** extension.

Load Dim_Customer

1. Go to File > New > Project, select Integration Services Project, name it `OLIST_ETL`, and click Create.
2. In the bottom Connection Managers area:
 - o Right-click > New OLE DB Connection....
 - o Add a connection pointing to your `OLIST_OLTP` database (Source).
 - o Add a second connection pointing to your `OLIST_DW` database (Destination).
3. Drag a Data Flow Task onto the canvas and rename it `Load_Dim_Customer`. Double-click to open it.
4. Drag an OLE DB Source into the Data Flow. Double-click it:
 - o Connection Manager: Select `OLIST_OLTP`.
 - o Data access mode: `Table or view`.
 - o Name of the table: `olist_customers_dataset`. Click OK.
5. Drag an OLE DB Destination into the Data Flow.
6. Connect the blue arrow from the Source to the Destination.
7. Double-click OLE DB Destination:
 - o Connection Manager: Select `OLIST_DW`.
 - o Table: `Dim_Customer`.
 - o Click the Mappings tab on the left. Connect:
 - o `customer_id` -> `CustomerBK`
 - o `customer_unique_id` -> `CustomerUniqueId`
 - o `customer_zip_code_prefix` -> `CustomerZipCode`
 - o `customer_city` -> `CustomerCity`
 - o `customer_state` -> `CustomerState`
 - o Click OK.

5.& 6.



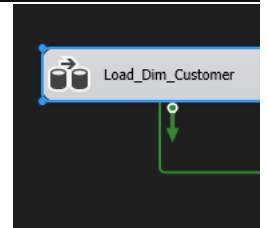
This instruction tells you how to set up the specific engine that v from your operational database into your data warehouse.

Here is exactly what it means and how to do it in Visual Studio:

- o "Drag a Data Flow Task onto the canvas": Look at the left screen for a panel called the SSIS Toolbox. Inside that tool Flow Task. Click it, hold your mouse button down, and drag space in the middle of your screen (the "canvas").
- o "Rename it `Load_Dim_Customer`": Once it is sitting on the name like "Data Flow Task 1". Right-click the box, select Rename `Load_Dim_Customer`. This is just a label so you know exactly what project does.
- o "Double-click to open it": Double-clicking the box will switch the overview tab (Control Flow) to a new, empty tab (Data Flow folder—you are now "inside" this specific task, ready to drag destination connections for the customer data.

3.

4.

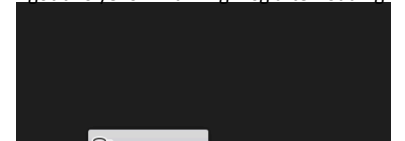


1. Find the OLE DB Source:
 - o Look at the SSIS Toolbox on the left side of your screen.  PNG
 - o Towards the bottom of that list, find **Other Sources**.  PNG
 - o Click the little arrow next to **Other Sources** to expand it.  PNG
 - o Inside, locate **OLE DB Source**.
2. Place it on the Canvas:
 - o Click, hold, and drag **OLE DB Source** into the large empty space on the screen.  PNG
3. Configure the Source:
 - o Double-click the **OLE DB Source** box that is now on your canvas; a configuration window will pop up.
 - o **OLE DB connection manager:** In the dropdown, select your `CHAMODDEALWIS.OLIST_OLTP`.  PNG
 - o **Data access mode:** Leave or set this to `Table or view`.
 - o **Name of the table or the view:** Open the dropdown and select `[dbo].[olist_customers_dataset]`.
 - o Click OK at the bottom of the window to save.

Configure it by following these steps:

1. Double-click the **OLE DB Destination** box on your canvas.  PNG
2. In the configuration window:
 - o **OLE DB connection manager:** Select `CHAMODDEALWIS.OLIST_DW` from the dropdown.  PNG
 - o **Data access mode:** Select `Table or view - fast load` (or `Table or view`).
 - o **Name of the table or the view:** Select `[dbo].[Dim_Customer]`.

I got this yellow warning msg after loading:



3. In the left menu of that window, click **Mappings**.
 4. Align your input columns to your destination columns:
 - `customer_id` → `CustomerBK`
 - `customer_unique_id` → `CustomerUniqueId`
 - `customer_zip_code_prefix` → `CustomerZipCode`
 - `customer_city` → `CustomerCity`
 - `customer_state` → `CustomerState`
 - Leave `CustomerSK` **unmapped** / `<ignore>`, because SQL Server generates this identity key automatically.
 5. Click **OK**.
- Once saved, the red X will disappear, confirming the task is properly configured. PNG

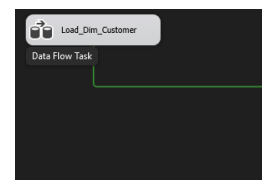


Load Dim_Product

1. Link the Tasks in the Control Flow

- Look at the top of your main visual canvas and click the **Control Flow** tab to zoom back out to your big-picture workflow.
- Open your **SSIS Toolbox** on the left, drag a new **Data Flow Task** onto the canvas, and rename it `Load_Dim_Product`.
- Click your existing `Load_Dim_Customer` task. You will see a green arrow pointing down. Click and drag that green arrow to connect it to your new `Load_Dim_Product` task. This ensures your customer data loads first, followed by your product data.

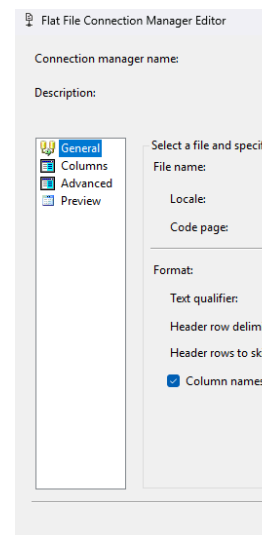
1. →



2. Configure the Flat File Source

- Double-click `Load_Dim_Product` to open its Data Flow canvas.
- From the **SSIS Toolbox** (under *Other Sources*), drag a **Flat File Source** onto the canvas.
- Double-click it to open the editor. Next to the *Flat file connection manager* dropdown, click **New...**
- **Connection manager name:** Type `Products_CSV`.
- **File name:** Click **Browse...** and select your `olist_products_dataset.csv` file.
- Click the **Columns** tab on the left to verify the data appears correctly, then click **OK** twice to close out.

2. →



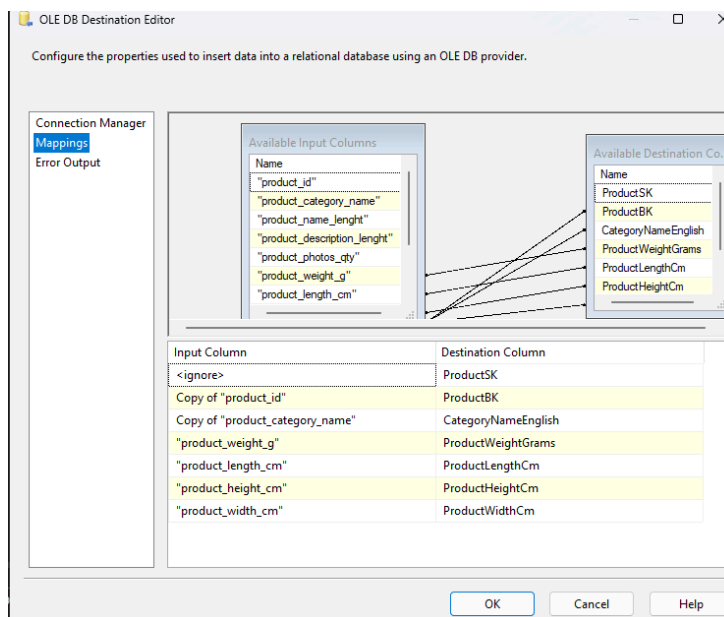
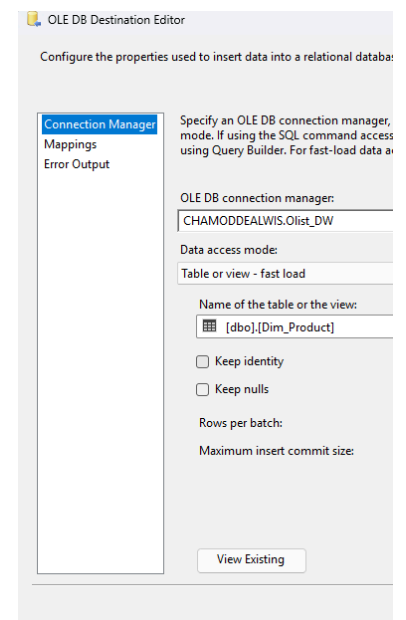
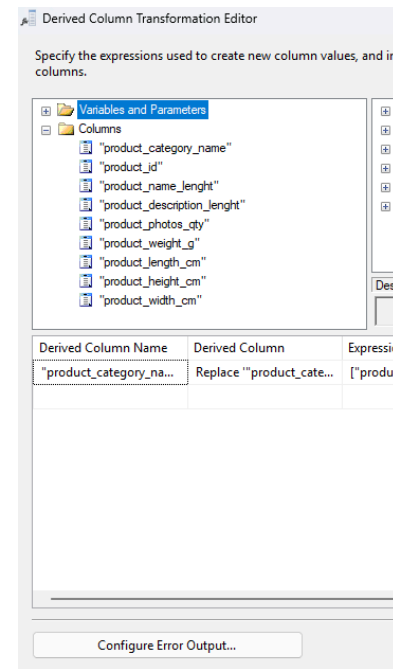
3. Handle Missing Values with a Derived Column

- From the **SSIS Toolbox** (under *Common*), drag a **Derived Column** transformation and place it below your Flat File Source.
- Connect the blue arrow from your Flat File Source down to the Derived Column.
- Double-click the **Derived Column** to open its editor.

- o In the bottom grid, under **Derived Column Name**, select `product_category_name` from the dropdown list.
- o Under **Derived Column**, ensure it says **Replace 'product_category_name'**.
- o Under **Expression**, type the following formula exactly to replace empty categories with the word "Unknown":
`REPLACENULL(product_category_name, "Unknown")`
- o Click OK to save.

4. Load the Data into the Destination

- o From the **SSIS Toolbox** (under *Other Destinations*), drag an **OLE DB Destination** onto the canvas below the Derived Column.
- o Connect the blue arrow from the Derived Column down to the OLE DB Destination.
- o Double-click the OLE DB Destination.
- o **Connection manager:** Select `CHAMODDEALWIS.Olist_DW`.
- o **Name of the table or view:** Select `[dbo].[Dim_Product]`.
- o Click **Mappings** on the left and connect the input columns to your destination table:
 - o `product_id` → `ProductBK`
 - o `product_category_name` → `CategoryNameEnglish`
 - o `product_weight_g` → `ProductWeightGrams`
 - o `product_length_cm` → `ProductLengthCm`
 - o `product_height_cm` → `ProductHeightCm`
 - o `product_width_cm` → `ProductWidthCm`
- o Click OK to finish.



Load fact table

1. Link the Control Flow

1. Return to the **Control Flow** tab.
2. Drag a new **Data Flow Task** onto the canvas and rename it `Load_Fact_Orders`.
3. Connect the green arrow from your previous task (`Load_Dim_Product`) down to `Load_Fact_Orders` so they run in the correct sequence.
4. Double-click `Load_Fact_Orders` to open its canvas.

2. Configure the OLE DB Source (The SQL Query)

1. Drag an **OLE DB Source** onto the canvas and double-click it.
2. **OLE DB connection manager:** Select `CHAMODDEALWIS.Olist_OLTP`.
3. **Data access mode:** Change this dropdown from *Table or view* to **SQL command**.
4. **SQL command text:** Paste your exact query into the large white box:

```
SELECT
o.order_id,
oi.order_item_id,
o.customer_id,
oi.product_id,
oi.seller_id,
CONVERT(INT, CONVERT(VARCHAR(8), o.order_purchase_timestamp, 112)) AS DateKey,
oi.price,
oi.freight_value,
(oi.price + oi.freight_value) AS TotalOrderValue,
DATEDIFF(day, o.order_purchase_timestamp, o.order_delivered_customer_date) AS
DeliveryTimeDays
FROM olist_orders_dataset o
JOIN olist_order_items_dataset oi ON o.order_id = oi.order_id
WHERE o.order_status = 'delivered';
```

OLE DB Source Editor

Configure the properties used by a data flow to obtain data from .

Connection Manager

Columns

Error Output

Specify an OLE DB connection manager mode. If using the SQL command access mode, using Query Builder.

OLE DB connection manager:

CHAMODDEALWIS.Olist_OLTP

Data access mode:

SQL command

SQL command text:

```
SELECT
o.order_id,
oi.order_item_id,
o.customer_id,
oi.product_id,
oi.seller_id,
CONVERT(INT, CONVERT(VARCHAR(8), o.order_purchase_timestamp, 112)) AS DateKey,
oi.price,
oi.freight_value,
(oi.price + oi.freight_value) AS TotalOrderValue,
DATEDIFF(day, o.order_purchase_timestamp, o.order_delivered_customer_date) AS
DeliveryTimeDays
FROM olist_orders_dataset o
```

Preview...

3. Add the Lookup Transformations

You will need three sequential Lookups to swap the alphanumeric IDs for your newly generated integer Surrogate Keys (SKs).

Lookup 1: Customer

1. Drag a **Lookup** transformation below your source and connect the blue arrow. Double-click it.
2. On the **General** tab, leave the defaults.
3. On the **Connection** tab, set the connection to `Olist_DW` and select the `[dbo].[Dim_Customer]` table.
4. On the **Columns** tab, drag `customer_id` from the Available Input Columns to `CustomerBK` in the Available Lookup Columns.
5. In the right-hand list, check the box next to `CustomerSK` to add it to your data flow. Click **OK**.

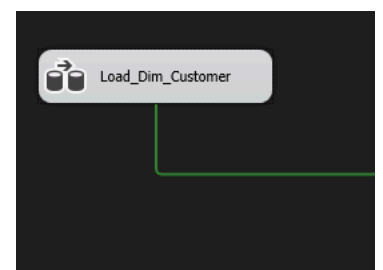
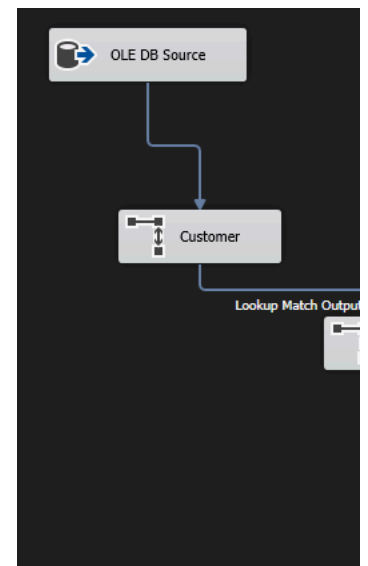
Lookup 2: Product

1. Drag a second **Lookup** below the first. Connect the blue arrow (select **Lookup Match Output** if prompted). Double-click it.
2. On the **Connection** tab, connect to `Olist_DW` and select `[dbo].[Dim_Product]`.
3. On the **Columns** tab, drag `product_id` to `ProductBK`.
4. Check the box next to `ProductSK`. Click **OK**.

Lookup 3: Seller

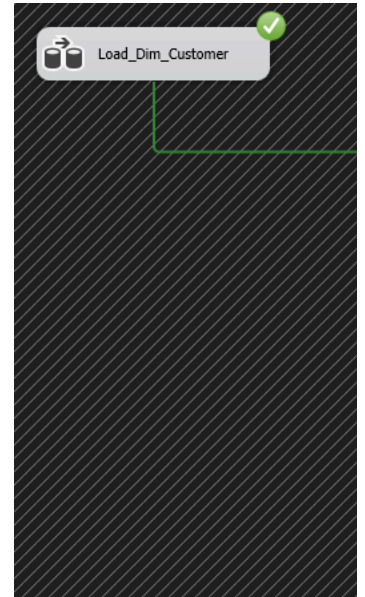
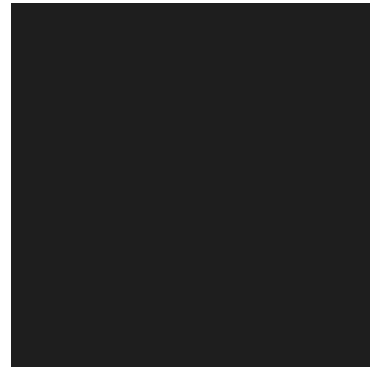
1. Drag a third **Lookup** below the second. Connect the blue arrow (Lookup Match Output). Double-click it.
2. On the **Connection** tab, connect to `Olist_DW` and select `[dbo].[Dim_Seller]`.
3. On the **Columns** tab, drag `seller_id` to `SellerBK`.
4. Check the box next to `SellerSK`. Click **OK**.

Results



4. Configure the Destination

1. Drag an **OLE DB Destination** to the bottom and connect the final blue arrow (Lookup Match Output) from the Seller Lookup.
2. Double-click it and set the connection to `OLIST_DW` and the table to `[dbo].[Fact_Orders]`.
3. Click **Mappings** and align the columns. Ensure your new keys map correctly:
 - `CustomerSK` → `CustomerSK`
 - `ProductSK` → `ProductSK`
 - `SellerSK` → `SellerSK`
 - Map `DateKey`, `order_id`, `price`, `freight_value`, `TotalOrderValue`, and `DeliveryTimeDays` to their matching destination columns.
4. Click **OK**.



5. Execute the Pipeline

With the Fact table configured, return to the **Control Flow** tab to see your complete three-step pipeline.

Press **F5** on your keyboard (or click the green **Start** button at the top of Visual Studio). The package will compile and run. If everything is configured correctly, you will see green checkmarks appear on all your tasks sequentially, indicating that your entire Data Warehouse is successfully populated and ready for Power BI!

Error handling

The log explicitly states that the "Seller" component failed because a "Row yielded no match during lookup". This happens when SSIS tries to find a `seller_id` in your `Dim_Seller` table, comes up empty, and crashes the pipeline as a safety measure. 

Here is the exact way to fix this so the pipeline finishes running:

1. **Stop Debugging:** Click the red square **Stop** button at the top of Visual Studio, or click the blue "switch to design mode" link at the bottom. 
2. **Open the Fact Task:** Double-click your `Load_Fact_Orders` task to view the pipeline.
3. **Edit the Lookups:** Double-click your Seller Lookup component.
4. **Change Error Handling:** On the **General** tab, find the dropdown menu labeled **Specify how to handle rows with no matching entries**.
5. Change it from *Fail component* to *Ignore failure*. (This tells SSIS to simply insert a `NULL` for the missing key instead of crashing).
6. Click **OK**.

Data Marts

```

USE Olist_DW;
GO

-- 1. Create a dedicated schema for the Logistics department
CREATE SCHEMA Logistics;
GO

-- 2. Create a targeted Data Mart view for shipping operations
CREATE VIEW Logistics.ShippingPerformance AS
SELECT
    f.OrderBK,
    d.FullDate AS PurchaseDate,
    c.CustomerCity AS DestinationCity,
    c.CustomerState AS DestinationState,
    s.SellerCity AS OriginCity,
    s.SellerState AS OriginState,
    f.FreightValue,
    f.DeliveryTimeDays
FROM dbo.Fact_Orders f
JOIN dbo.Dim_Date d ON f.DateKey = d.DateKey
JOIN dbo.Dim_Customer c ON f.CustomerSK = c.CustomerSK
JOIN dbo.Dim_Seller s ON f.SellerSK = s.SellerSK;
GO

USE Olist_DW;
GO

-- =====
-- 1. Create the Schemas (Departments)
-- =====
CREATE SCHEMA Sales;
GO
CREATE SCHEMA Marketing;
GO
CREATE SCHEMA Executive;
GO

-- =====
-- 2. Sales Data Mart: Product Performance
-- Focuses on product categories and revenue
-- =====
CREATE VIEW Sales.ProductPerformance AS
SELECT
    p.CategoryNameEnglish,
    p.ProductBK AS ProductID,
    d.FullDate AS SaleDate,
    f.Price,
    f.TotalOrderValue
FROM dbo.Fact_Orders f
JOIN dbo.Dim_Product p ON f.ProductSK = p.ProductSK
JOIN dbo.Dim_Date d ON f.DateKey = d.DateKey;
GO

-- =====
-- 3. Marketing Data Mart: Customer Insights
-- Focuses on customer geography and seasonal buying
-- =====
CREATE VIEW Marketing.CustomerInsights AS
SELECT
    c.CustomerBK AS CustomerID,
    c.CustomerCity,
    c.CustomerState,
    d.Year,
    d.MonthName,
    d.FullDate AS OrderDate,
    f.TotalOrderValue
FROM dbo.Fact_Orders f
JOIN dbo.Dim_Customer c ON f.CustomerSK = c.CustomerSK
JOIN dbo.Dim_Date d ON f.DateKey = d.DateKey;
GO

-- =====
-- 4. Executive Data Mart: Monthly Summary
-- High-level aggregations of revenue and volume
-- =====
CREATE VIEW Executive.MonthlySummary AS
SELECT
    d.Year,
    d.Month,
    d.MonthName,
    COUNT(DISTINCT f.OrderBK) AS TotalOrders,
    SUM(f.TotalOrderValue) AS TotalRevenue,
    SUM(f.FreightValue) AS TotalFreightCosts
FROM dbo.Fact_Orders f
JOIN dbo.Dim_Date d ON f.DateKey = d.DateKey
GROUP BY
    d.Year,
    d.Month,
    d.MonthName;
GO

```

